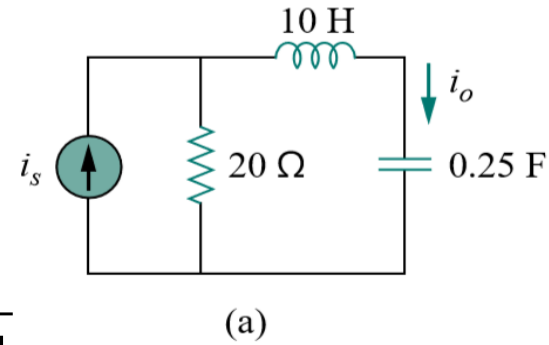


Solutions to Frequency Response

1. Obtain the transfer function $H(\omega) = I_o/I_s$.

(a) **Current division**

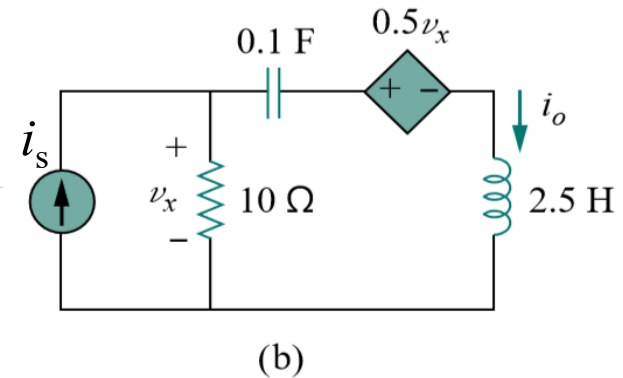
$$H_1(j\omega) = \frac{\dot{I}_0}{\dot{I}_s} = \frac{R}{R + j\omega L + \frac{1}{j\omega C}} = \frac{2j\omega}{(j\omega)^2 + 2j\omega + 0.4}$$



(b) **Mesh analysis**

$$\begin{cases} (10 + j\omega 2.5 + \frac{1}{j\omega 0.1})\dot{I}_0 - 10\dot{I}_s = -0.5\dot{V}_x \\ \dot{V}_x = 10(\dot{I}_s - \dot{I}_0) \end{cases}$$

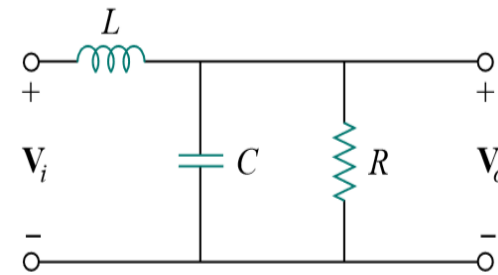
$$H_2(j\omega) = \frac{\dot{I}_0}{\dot{I}_s} = \frac{2j\omega}{(j\omega)^2 + 2j\omega + 4}$$



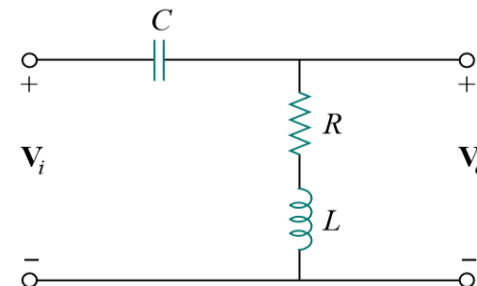
2. Find the transfer function $H(\omega) = V_o/V_i$

(a) Voltage division

$$H_1(j\omega) = \frac{\dot{V}_o}{\dot{V}_s} = \frac{\frac{1}{j\omega C + 1/R}}{j\omega L + \frac{1}{j\omega C + 1/R}} = \frac{1/LC}{(j\omega)^2 + j\omega/RC + 1}$$



(a)



(b)

(b) Voltage division

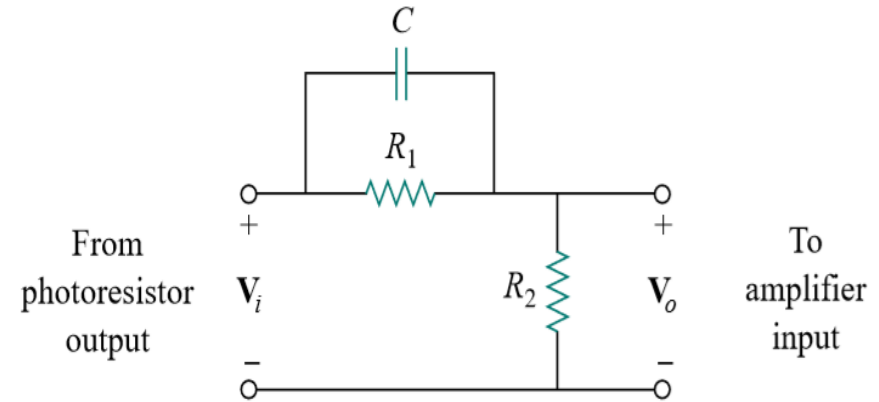
$$H_2(j\omega) = \frac{\dot{V}_o}{\dot{V}_s} = \frac{j\omega L + R}{j\omega L + R + \frac{1}{j\omega C}} = \frac{(j\omega)^2 + j\omega R/L}{(j\omega)^2 + j\omega R/L + 1/LC}$$

3. The RC circuit in Fig. 3 is used for a lead compensator in a system design. Obtain the transfer function of the circuit.

Voltage division

$$H_1(j\omega) = \frac{\dot{V}_0}{\dot{V}_s} = \frac{R_2}{R_2 + \frac{1}{j\omega C + 1/R_1}}$$

$$= \frac{j\omega R_2 C + \frac{R_2}{R_1}}{j\omega R_2 C + \frac{R_2}{R_1} + 1} = \frac{j\omega + \frac{1}{R_1 C}}{j\omega + \frac{1}{R_1 C} + \frac{1}{R_2 C}}$$

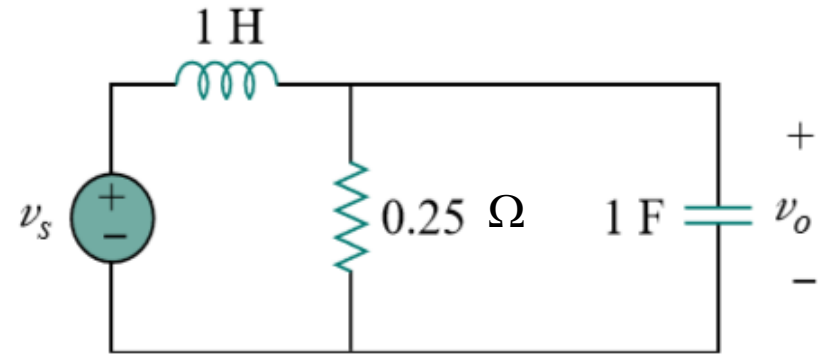


5. Find the transfer function V_o/V_s of the circuit in Fig. 5. Show that the circuit is a low-pass filter.

$$H(j\omega) = \frac{\dot{V}_o}{\dot{V}_s} = \frac{\frac{1}{j\omega C + 1/R}}{j\omega L + \frac{1}{j\omega C + 1/R}}$$

$$= \frac{1/LC}{(j\omega)^2 + j\omega/RC + 1}$$

$$= \frac{1}{(j\omega)^2 + 4j\omega + 1}$$



6. In a high-pass RL filter with a cutoff frequency of 100 kHz, $L = 40$ mH. Find R .

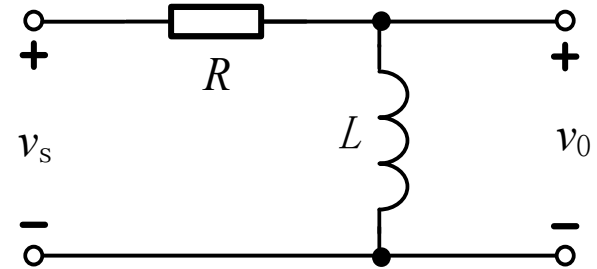
$$H(\omega) = \frac{j\omega L}{R + j\omega L} = \frac{j\omega}{j\omega + R/L}$$

$$= \frac{j\frac{\omega}{\omega_c}}{1 + j\frac{\omega}{\omega_c}}$$

$$\omega_c = \frac{R}{L}$$

$$2\pi \times 100 \times 10^3 = \frac{R}{0.04}$$

$$R = 25\text{k}\Omega$$



7. Determine the center frequency and bandwidth of the band-pass filters in Fig. 7.

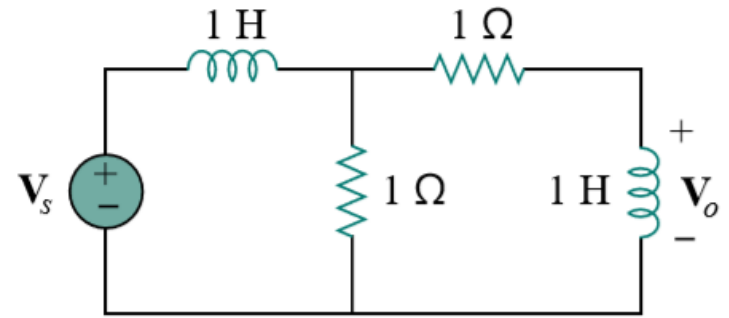
$$H(j\omega) = \frac{\dot{V}_o}{\dot{V}_s} = \frac{\frac{j\omega + 1}{j\omega + 1 + 1}}{j\omega + \frac{j\omega + 1}{j\omega + 1 + 1}} \frac{j\omega}{j\omega + 1}$$

$$= \frac{j\omega}{(j\omega)^2 + 3j\omega + 1}$$

$$\omega = 0, H = 0$$

$$\omega \rightarrow \infty, H \rightarrow 0$$

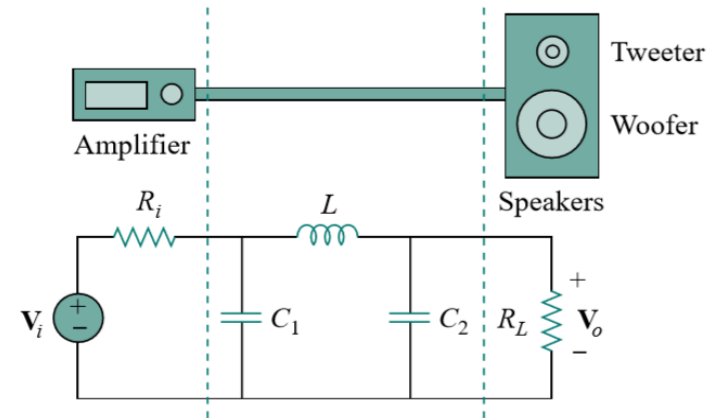
So it is a band-pass filter.



$$\omega_0 = 1\text{ rad/s}$$

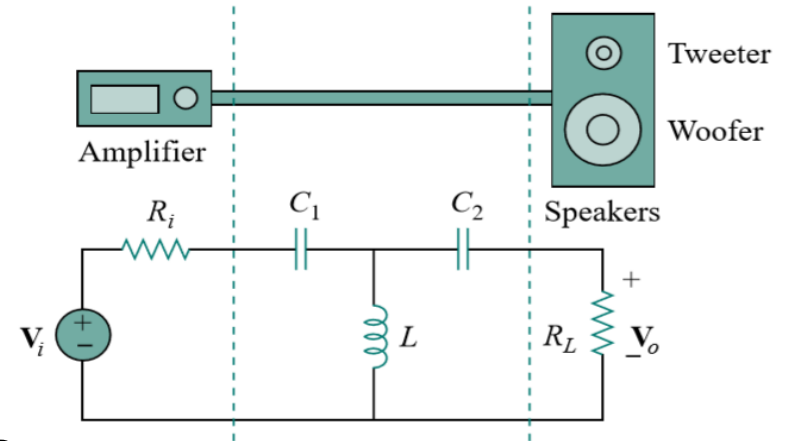
$$B = \frac{\omega_0}{Q} = 3\text{ rad/s}$$

8. The crossover circuit in Fig. 8 is a low-pass filter that is connected to a woofer. Find the transfer function $H(\omega) = V_o(\omega)/V_i(\omega)$.



$$H = \frac{\dot{V}_o}{\dot{V}_s} = \frac{Z_{C_1} // [Z_L + (Z_{C_2} // R_L)]}{R_i + Z_{C_1} // [Z_L + (Z_{C_2} // R_L)]} \frac{(Z_{C_2} // R_L)}{Z_L + (Z_{C_2} // R_L)}$$

9. The crossover circuit in Fig. 9 is a high-pass filter that is connected to a tweeter. Determine the transfer function $H(\omega) = V_o(\omega)/V_i(\omega)$.

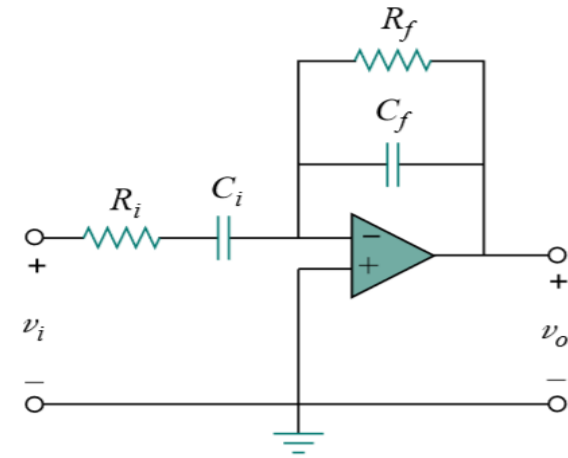


$$H = \frac{\dot{V}_o}{\dot{V}_s} = \frac{Z_L \parallel (Z_{C_2} + R_L)}{R_i + Z_{C_1} + Z_L \parallel (Z_{C_2} + R_L)} \frac{R_L}{(Z_{C_2} + R_L)}$$

**11. Obtain the transfer function of the active filter in Fig. 11.
What kind of filter is it?**

Nodal analysis

$$\frac{\dot{V}_s}{R_i + \frac{1}{j\omega C_i}} = -\left(\frac{1}{R_f} + j\omega C_f\right)\dot{V}_o$$

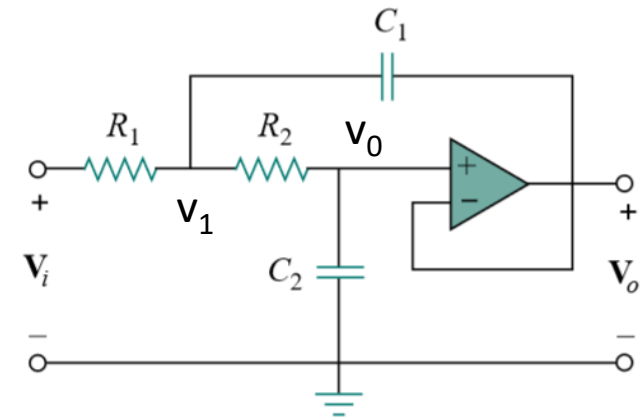


$$H(j\omega) = \frac{\dot{V}_o}{\dot{V}_s} = \frac{-j\omega \frac{1}{C_f R_i}}{(j\omega)^2 + j\omega\left(\frac{1}{C_i R_i} + \frac{1}{C_f R_f}\right) + \frac{1}{C_i C_f R_f R_i}}$$

Band-pass

14. A second-order active filter known as a Butterworth filter is shown in Fig. 14.

- (a) Find the transfer function V_o/V_i .**
(b) Show that it is a low-pass filter.



Nodal analysis

$$\left(\frac{1}{R_1} + \frac{1}{R_2} + j\omega C_1\right)V_1 - \frac{1}{R_1}V_i - \frac{1}{R_2}V_o - j\omega C_1V_o = 0$$

$$\left(\frac{1}{R_2} + j\omega C_2\right)V_o - \frac{1}{R_2}V_1 = 0$$

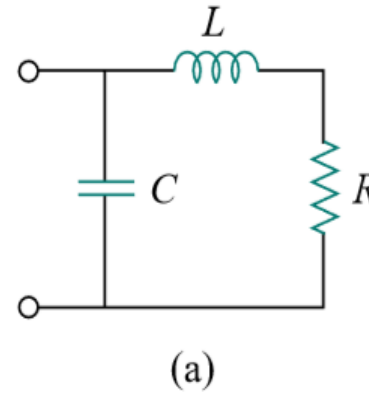
$$\omega = 0, H = 1$$

$$\omega \rightarrow \infty, H \rightarrow 0$$

$$H(j\omega) = \frac{V_o}{V_i} = \frac{1 / C_1 C_2 R_1 R_2}{(j\omega)^2 + j\omega \frac{R_1 + R_2}{C_1 R_1 R_2} + \frac{1}{C_1 C_2 R_1 R_2}}$$

LOWPASS

15. Calculate the resonant frequency of each of the circuits in Fig.15.



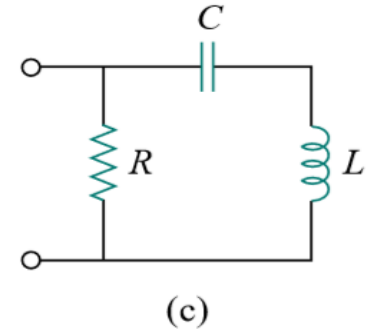
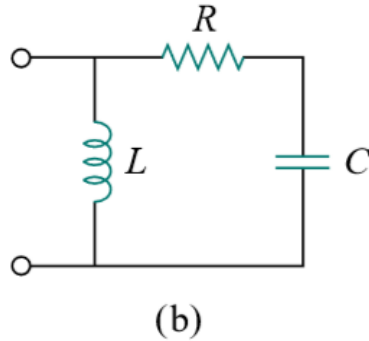
$$Y = j\omega C + \frac{1}{R + j\omega L} = j\omega C + \frac{R - j\omega L}{R^2 + \omega^2 L^2}$$

$$\text{Im}[Y] = \omega C - \frac{\omega L}{R^2 + \omega^2 L^2} = 0$$

$$\omega_0^2 = \frac{-R^2 C + L}{L^2 C}$$

$$\omega_0 = \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

15. Calculate the resonant frequency of each of the circuits in Fig.15.



$$Y = \frac{1}{j\omega L} + \frac{1}{R + 1/j\omega C} = -j\frac{1}{\omega L} + \frac{R - 1/j\omega C}{R^2 + 1/\omega^2 C^2}$$

$$Z_{LC} = 1/j\omega C + j\omega L = j(\omega L - \frac{1}{\omega C})$$

$$\text{Im}[Y] = -\frac{1}{\omega L} + \frac{1/\omega C}{R^2 + 1/\omega^2 C^2} = 0$$

$$\text{Im}[Z_{LC}] = j(\omega L - \frac{1}{\omega C}) = 0$$

$$\frac{1}{\omega_0^2} = LC - R^2 C^2$$

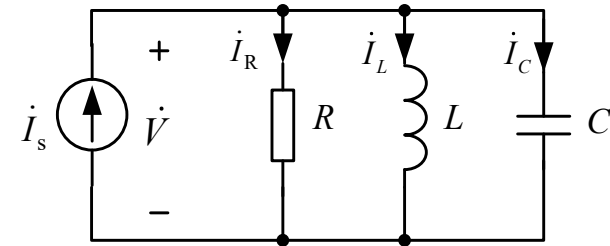
$$\omega_0 = 1/\sqrt{LC}$$

$$\omega_0 = \sqrt{\frac{1}{LC - R^2 C^2}}$$

17. It is expected that a parallel RLC resonant circuit has a mid-band admittance of $25 \times 10^{-3} \text{ S}$, quality factor of 80, and a resonant frequency of 200 k rad/s. Calculate the values of R , L , and C . Find the bandwidth and the half-power frequencies.

$$Y(\omega_0) = 25 \times 10^{-3} \text{ S} \quad R = 1 / 25 \times 10^{-3} = 40 \Omega$$

$$Q = 80 \quad \omega_0 = 200 \times 10^3 \text{ rad / s}$$



$$Q = \frac{R}{\omega_0 L} = 80, \quad L = \frac{R}{80 \omega_0} = 2.5 \mu\text{H}$$

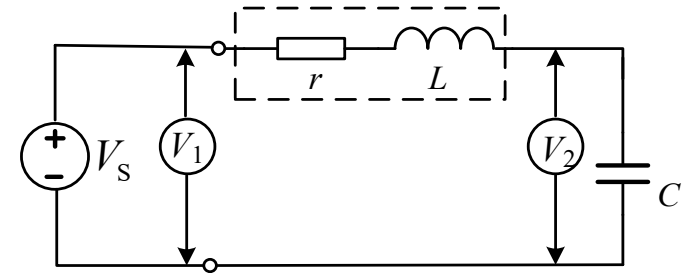
$$Q = \omega_0 C R = 80, \quad C = \frac{80}{\omega_0 R} = 10 \mu\text{F}$$

$$B = \frac{\omega_0}{Q} = \frac{200 \text{ k}}{80} = 2.5 \text{ k rad/s}$$

$$\text{high } Q, \quad \omega_c = \omega_0 \pm \frac{B}{2} = 198.75\text{-}201.25 \text{ k rad/s}$$

19. The dotted portion in the circuit shown in Fig.19 is the equivalent circuit of a practical inductor. V_1 and V_2 are AC voltmeters. When the frequency of the voltage source is adjusted to $f=900\text{kHz}$, voltmeter V_2 shows the maximum value and $V_1=1\text{mV}$, $V_2=62\text{mV}$. Given $C=200\text{pF}$, find L and r .

When voltmeter V_2 shows the maximum value, the circuit is resonant, so



$$\omega_0 = \frac{1}{\sqrt{LC}} = 2\pi f_0, L = \frac{1}{(2\pi f_0)^2 C} = \frac{1}{648\pi^2} \approx 1.56 \times 10^{-4} \text{ H}$$

$$V_1 = V_s = 1\text{mV}, V_2 = QV_s = 62\text{mV}, Q=62 = \frac{\omega_0 L}{r}$$

$$r = \frac{\omega_0 L}{62} \approx 14.22\Omega$$

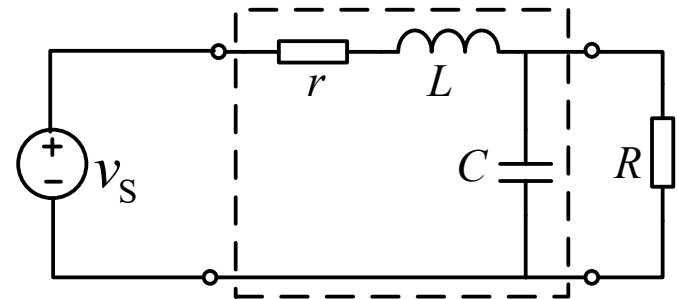
20. In the series resonant circuit shown in Fig. 20(between the four little circles), given characteristic impedance $\rho = 1\text{k}\Omega$, there is no loss in capacitor. The inductor has an equivalent loss resistance r and its quality factor is $Q_L = 100$. The resonant angular frequency is 100k rad/s . $R = 50\text{k}\Omega$ is the load connected to the circuit. Find the quality factor Q of the circuit, bandwidth, L and C ; If the maximum value of V_s is 1V , considering the possibility of load R 's disconnection to the circuit, find the minimum withstand voltage for the capacitor C .

$$r = \frac{\rho}{Q_L} = \frac{1000}{100} = 10 \Omega$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = 100\text{k}$$

$$\rho = \sqrt{L/C} = 1\text{k}$$

$$C = 10\text{nF}, L = 10\text{mH}$$



$$\omega_0 = 100 \text{krad/s} \quad C = 10 \text{nF}, L = 10 \text{mH}$$

Applying parallel model for Q_C :

$$Q_C = \omega_0 CR = 100 \text{k} \times 10 \text{n} \times 50 \text{k} = 50$$

Convert parallel model to series model:

$$r_C = \frac{X_C^2}{R} = \frac{\rho^2}{R} = \frac{(1 \times 10^3)^2}{50 \times 10^3} = 20 \, \Omega$$

Quality factor in on-load circuit:

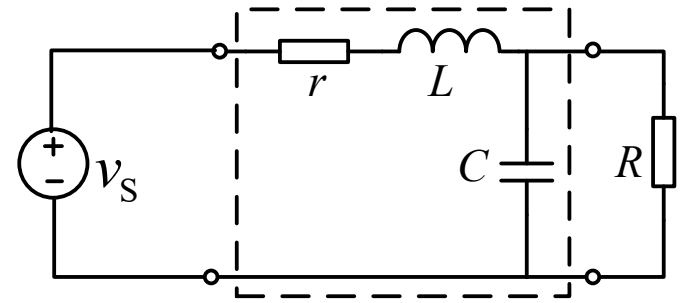
$$Q' = \frac{\rho}{r_L + r_C} = \frac{1 \times 10^3}{10 + 20} = 33.3$$

When $V_{s_m} = 1 \text{V}$, R disconnects

$$Q = Q_L = 100$$

$$V_{C0} = QV_S = 100 \text{ V}$$

The minimum withstand voltage $V_C = 100 \text{V}$



21. The equivalent circuit of an intermediate frequency amplifier is shown in Figure 21. Given the internal resistance of the source $R_s=150\text{k}\Omega$, inductance $L=590\mu\text{H}$, resistance $r=14.3\Omega$, capacitance $C=200\text{pF}$ and its loss neglected, and the equivalent load resistance $R_L=500\text{k}\Omega$, find the resonant frequency and the bandwidth.

$$Q_L = \frac{\rho}{r} = \frac{\sqrt{L/C}}{r}$$

$$= \frac{\sqrt{590 \times 10^{-6} / 200 \times 10^{-12}}}{14.3} = 120$$

$$R \approx Q_L^2 r = 120^2 \times 14.3 = 205.92\text{k}\Omega$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = 2.9111\text{Mrad/s} \quad R_p = 1 / (1/150 + 1/500 + 1/205.9) = 74\text{k}\Omega$$

$$Q = \frac{R_p}{\omega_0 L} = \frac{74\text{k}}{2.91\text{M} \times 590\mu} = 43.1 \quad B = \frac{\omega_0}{Q} = \frac{2.91\text{M}}{43.1} = 67.6\text{k rad/s}$$

$$\text{high } Q, \quad \omega_c = \omega_0 \pm \frac{B}{2} = 2877.3\text{k} - 2944.9\text{k rad/s}$$

